CLIENT REPORT

NYC-311

Improve understanding of city service demand and support more efficient resource allocation.

REPORT REFERENCE

AUD-8663C7D0D4

GENERATION DATE

26 August 2026

BÓVEDA VERSION

Alpha 1.0.2

Generated from the available project record

CONTENTS

In this report

The project and Bóveda's position

A concise view of the project, its primary recorded result and the parts of the record that need attention.

What the project does

The project examines NYC 311 demand, complaint patterns, resolution times and storm-related differences, using descriptive analysis, classification and time-series forecasting. Its outputs are intended to support city resource allocation and resident expectations, although the evaluated resolution model is restricted to Brooklyn heat and hot-water complaints.

Evidence E-B335630551 · E-B9CC546808 · E-2B1FA53FA7

Best recorded result

Test accuracy

0.6248641707003852

Model

Random Forest Classifier with 100 trees

Target

Three resolution-time classes for Brooklyn heat complaints

Evaluation

Held-out X_test/Y_test from filtered 2015 model data

Evidence E-B9CC546808 · E-2B1FA53FA7

Partial reconstruction

Bóveda could trace the main result to the data, model and evaluation sample behind it. However, one important applicable analytical area could not be reconstructed, so the overall reconstruction remains partial.

What needs attention

1	4
Signal	Evidence Gaps
1 High	

Overall evidence position

The available project record supports some areas more completely than others, but exact reproduction remains incomplete because 4 evidence gaps are still unresolved.

Coverage not established

PROJECT EXPLANATION

Project overview

What the project is intended to do, the objects it analyses and the use it is intended to support.

Purpose

The project examines NYC 311 demand, complaint patterns, resolution times and storm-related differences, using descriptive analysis, classification and time-series forecasting. Its outputs are intended to support city resource allocation and resident expectations, although the evaluated resolution model is restricted to Brooklyn heat and hot-water complaints.

Evidence E-B335630551 · E-B9CC546808 · E-2B1FA53FA7

Analytical task	Describe request patterns, classify resolution-time ranges, forecast daily volumes, and compare storm-related response times.

	OBSERVED Evidence E-B335630551 · E-B9CC546808
Target / outcome	Outputs a three-class complaint-resolution range and forecast daily service-request volume. OBSERVED Evidence E-B9CC546808
Unit of observation	One individual 311 service request INTERPRETED_INFERRED Evidence E-B9CC546808 · E-2B1FA53FA7
Population / scope	NYC 311 service requests; resolution modelling narrows to Brooklyn heat and hot-water complaints. INTERPRETED_INFERRED Evidence E-B9CC546808 · E-2B1FA53FA7
Intended use	Support city resource allocation and help residents estimate when complaints may be resolved. OBSERVED Evidence E-B9CC546808
Project period	The prepared service-request analysis and resolution model used data from 2015. INTERPRETED_INFERRED Evidence E-2B1FA53FA7
Major data sources	NYC Open Data 311 Service Requests dataset. OBSERVED Evidence E-B335630551 · E-B9CC546808

DATA & POPULATIONS

What data the project operated on

Only supervisory questions answered by the available evidence are included below.

Data in brief

Public NYC 311 records were prepared with created and closed dates, location, complaint descriptors and derived resolution days; a filtered Brooklyn heat subset fed classification.

Evidence E-B9CC546808 · E-2B1FA53FA7

Source data	More than 16M rows OBSERVED Evidence E-B9CC546808
Model sample	47,239 observations DERIVED Evidence E-2B1FA53FA7
Evaluation sample	20,246 observations DERIVED Evidence E-2B1FA53FA7
Main population filters	Resolution modelling selected Brooklyn heat and hot-water complaints and removed requests with resolution times exceeding 28 days. OBSERVED Evidence E-2B1FA53FA7
Known population limitation	The resolution model is limited to Brooklyn heat and hot-water complaints resolved within 28 days, restricting application beyond that selected group. INTERPRETED_INFERRED Evidence E-2B1FA53FA7

Original data **16,000,000**

source

Prepared population **2,099,575**

population stage

Prepared population **654,694**

population stage

Prepared population **67,497**

population stage

Filtered model population **67,495**

population stage

Filtered model population **67,485**

population stage

Training data **47,239**

training

Evaluation data **20,246**

evaluation

Missing data before preparation

Quantified missingness for variables connected to the reconstructed analysis.

VARIABLE	MISSINGNESS	DENOMINATOR	EVIDENCE
Descriptor	0%	67,497	DERIVED Evidence E-2B1FA53FA7
Incident Zip	0.003%	67,497	DERIVED Evidence E-2B1FA53FA7
Day of Week	0%	67,497	DERIVED Evidence E-2B1FA53FA7
Day of Month	0%	67,497	DERIVED Evidence E-2B1FA53FA7
Month	0%	67,497	DERIVED Evidence E-2B1FA53FA7

Population / sample lineage

How the original population became the training and evaluation samples.

Original data

→ filter →

Prepared population

Resolution modelling selected Brooklyn heat and hot-water complaints and removed requests with resolution times exceeding 28 days.

Evidence E-B9CC546808 · E-2B1FA53FA7

Prepared population

→ filter →

Prepared population

dfBrooklyn.shape.

Evidence E-2B1FA53FA7

Prepared population

→ filter →

Prepared population

getDfSummary(dfBrookHeat).

Evidence E-2B1FA53FA7

Prepared population

→ filter →

Filtered model population

dfmodel = buildFeatures(dfBrookHeat) dfmodel = dfmodel[['Incident Zip','Day of Week','Day of Month','Month','descriptorAPARTMENT ONLY', 'descriptorENTIRE BUILDING','ResolutionTime']] dfmodel.describe().

Evidence E-2B1FA53FA7

Filtered model population

→ filter →

Filtered model population

bins = [0,2,6,28] groupnames = [0,1,2] dfmodel['categories'] = pd.cut(dfmodel['ResolutionTime'], bins,includelowest=True,labels=groupnames) dfmodel.describe().

Evidence E-2B1FA53FA7

Filtered model population

➝ split ➝

Training data

70/30 train/test split.

Evidence E-2B1FA53FA7

Filtered model population

➝ split ➝

Evaluation data

70/30 train/test split.

Evidence E-2B1FA53FA7

What was recorded, and what it supports

The strongest comparable final result is shown with its evaluation context and material limitation.

Best recorded result

Test accuracy

0.6248641707003852

Model
Random Forest Classifier with 100 trees

Target
Three resolution-time classes for Brooklyn heat complaints

Evaluation sample
Held-out X_test/Y_test from filtered 2015 model data

Evidence E-B9CC546808 · E-2B1FA53FA7

Evaluation design

A 70/30 train/test split produced 47,239 training and 20,246 evaluation observations.

DERIVED

Evidence E-2B1FA53FA7

Principal limitation

The three classes are imbalanced, with too few examples in the longest-resolution class, limiting interpretation of overall accuracy.

OBSERVED

Evidence E-B9CC546808 · E-2B1FA53FA7

Other recorded material results

These outcomes retain their own target and evaluation context. They are not treated as directly comparable unless a model comparison below establishes a shared setup.

RESULT	METHOD / TARGET	EVALUATION CONTEXT	EVIDENCE
r2_score 0.953888	Prophet additive time-series model Log daily NYC 311 request volume	Goodness of fit on 391 retained days with over 25 calls	OBSERVED Evidence E-B9CC546808 · E-2B1FA53FA7

What the result establishes

On the recorded test dataset, the random forest achieved accuracy of 0.62486417070038525 for classifying complaint resolution-time ranges.

Evidence E-2B1FA53FA7

What it does not establish

It does not establish similar accuracy for other complaint types, boroughs, periods, or requests taking more than 28 days.

Evidence E-B9CC546808 · E-2B1FA53FA7

Model comparison

Three resolution-time classes for Brooklyn heat complaints

Results recorded for the same target and evaluation setup.

METRIC	LOGISTIC REGRESSION	DECISION TREE	RANDOM FOREST
Accuracy calculated from yes/no predictions	0.5780894991603279	0.621456090092	0.6248641707003852

Random Forest records the strongest value across all 1 comparable metrics shown for Three resolution-time classes for Brooklyn heat complaints. The comparison is still row-specific: the values describe different evaluation measures and should not be compared vertically as though they shared one scale.

Evaluation behaviour

Recorded diagnostics show correct classifications and errors that one headline score can hide.

Random Forest

Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data

Rows: actual · columns: predicted

	Predicted 0	Predicted 1
Actual 0	1	
Actual 1		

Random Forest correctly classifies 24.2% of actual more than a week observations, compared with 73.5% for 2 <= days <= 6. Its most common error for more than a week is assignment to 2 <= days <= 6 (53.5%).

Evidence E-2B1FA53FA7

What Bóveda found

Signals describe material project conditions detected by deterministic checks. They do not automatically mean errors, failures, violations or compliance problems.

01 High

Feature / driver evidence

Important analytical evidence is missing

What Bóveda found

An applicable analytical area is entirely unavailable because the project record does not contain enough evidence for meaningful supervisory assessment.

Why it matters

The missing evidence prevents Bóveda from meaningfully assessing an important part of the project, rather than merely limiting a detail within an otherwise assessable area.

Supported by 1 verification check

Evidence E-2B1FA53FA7 · E-B9CC546808

Where the record is incomplete

These are parts of the project that Bóveda could not fully verify or reproduce from the available evidence. A gap does not mean something is wrong; it means the project record is incomplete.

01 Evidence Gap

Reproducibility

The exact data version used for the result is not fully identified

What Bóveda found

Bóveda found the project data, but not enough information to identify the precise dataset snapshot behind the reported result.

Why it matters

A precise data version is needed to reproduce the same analysis from the same inputs.

Supported by 1 verification check

Evidence E-2B1FA53FA7

02 Evidence Gap

Reproducibility

Bóveda cannot fully link the predictions to the labels used to evaluate them

What Bóveda found

The available project record does not contain enough information to reconstruct the complete link between predictions and evaluation labels.

Why it matters

Without that link, Bóveda cannot fully reproduce or independently verify the reported evaluation result.

Supported by 1 verification check

Evidence E-2B1FA53FA7

03 Evidence Gap

Reproducibility

The exact software environment cannot be fully reconstructed

What Bóveda found

Some configuration or environment details needed to recreate the analysis are missing from the available project record.

Why it matters

Without these details, Bóveda cannot establish the complete execution setup behind the reported result.

Supported by 1 verification check

Evidence E-2B1FA53FA7

04 Evidence Gap

Feature / driver evidence

Feature and driver evidence cannot be assessed

What Bóveda found

Bóveda found a main model result, but no linked evidence showing which inputs the model used or considered important.

Why it matters

Without linked feature evidence, a supervisor cannot meaningfully assess what drove the displayed model result.

Supported by 1 verification check

Evidence E-2B1FA53FA7 · E-B9CC546808

PROJECT HISTORY

Material events in the available record

Bóveda found 15 recorded project changes, but none could be connected to a material data, model, evaluation, or release event. The remaining activity is retained as supporting context.

2015

data period

Project data period reconstructed

The prepared service-request analysis and resolution model used data from 2015.

This establishes the period represented by the project evidence; it does not establish when the dataset was created or changed.

Evidence E-2B1FA53FA7

2015

evaluation

Primary evaluation recorded

recorded project evidence records Test accuracy 0.62486417070038525 for Three resolution-time classes for Brooklyn heat complaints using Random Forest Classifier with 100 trees.

Evidence E-B9CC546808 · E-2B1FA53FA7

TRACEABILITY & EVIDENCE COVERAGE

How completely Bóveda reconstructed the project

Reconstruction confidence combines the evidence trail behind the main result with whether important applicable analytical areas could be reconstructed. It does not measure model quality, project quality, performance, compliance or governance.

Project reconstruction

Partial reconstruction

Bóveda could trace the main result to the data, model and evaluation sample behind it. However, one important applicable analytical area could not be reconstructed, so the overall reconstruction remains partial.

- ✓ The recorded result can be identified.
- ✓ The result is connected to its recorded metric and value.
- ✓ The result is connected to a model and target.
- ✓ The evaluation context can be traced.
- ✓ The evaluation sample and main result can be traced together.
- ✓ Findings retain links to verification checks and Evidence Tokens.

Evidence coverage

Coverage not established

Project Context
Sufficient
Whether the project purpose, scope, methods, decisions and limitations are supported.

Traceability
Not assessed
Whether the main conclusions link to identifiable source material.

Reproducibility
Partial
Whether the main analytical result can be recreated from the available project material.

Source
Sufficient
Whether the source material needed to support the main claims was available for review.

INTERPRETATION BOUNDARY

Remaining limitations

What the client should not conclude from this report, based on the material boundaries represented in the current project record.

01 The resolution model is limited to Brooklyn heat and hot-water complaints resolved within 28 days, restricting application beyond that selected group.

02 The three classes are imbalanced, with too few examples in the longest-resolution class, limiting interpretation of overall accuracy.

03 It does not establish similar accuracy for other complaint types, boroughs, periods, or requests taking more than 28 days.

04 Bóveda found the project data, but not enough information to identify the precise dataset snapshot behind the reported result. A precise data version is needed to reproduce the same analysis from the same inputs.

05 The available project record does not contain enough information to reconstruct the complete link between predictions and evaluation labels. Without that link, Bóveda cannot fully reproduce or independently verify the reported evaluation result.

06 Some configuration or environment details needed to recreate the analysis are missing from the available project record. Without these details, Bóveda cannot establish the complete execution setup behind the reported result.

07 Bóveda found a main model result, but no linked evidence showing which inputs the model used or considered important. Without linked feature evidence, a supervisor cannot meaningfully assess what drove the displayed model result.

Absence of evidence is not evidence of absence. These statements are limited to the available project record reviewed by Bóveda.

APPENDIX A

Technical appendix

Detailed reconstruction retained for technical review. Internal identities and precise analytical terminology are intentionally confined to this layer.

Complete target definitions

TARGET	SOURCE FIELD	CONSTRUCTION	POSITIVE CLASS	INTERPRETATION	STATUS & EVIDENCE
Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data	`categories`	`pd.cut(df_model['Resolution_Time'], bins,include_lowest=True,labels=group_names)`	–	derived binary target	OBSERVED E-2B1FA53FA7

Target-construction ancestry

STEP	TARGET	OPERATION	EXPRESSION / FIELD	STATUS & EVIDENCE
ATCS-F1371CB596	Target not established		`df_model['Resolution_Time'].astype(int)`	OBSERVED E-2B1FA53FA7
ATCS-4FB137B859	Target not established		`pd.cut(df_model['Resolution_Time'], bins,include_lowest=True,labels=group_names)`	OBSERVED E-2B1FA53FA7
ATCUNK-2D245BD468	Target not established		–	DERIVED E-2B1FA53FA7
ATCSRC-0A6536E809	Target not established		–	OBSERVED E-2B1FA53FA7
ATCUNK-1B6D222610	Target not established		–	DERIVED E-2B1FA53FA7
ATCUNK-2CE6A7CB87	Target not established		–	DERIVED E-2B1FA53FA7
ATCUNK-48FC4176AF	Target not established		–	DERIVED E-2B1FA53FA7
ATCUNK-78BAC9136F	Target not established		–	DERIVED E-2B1FA53FA7

Full population and sample table

POPULATION	ROLE	COUNT	UNIT	PREDICATE	STATUS & EVIDENCE
Prepared population	population stage	2,099,575	observations	`dfperfect = prepareData(df) dfperfect.shape`	OBSERVED E-2B1FA53FA7
Prepared population	population stage	654,694	observations	`dfBrooklyn.shape`	OBSERVED E-2B1FA53FA7
Prepared population	population stage	67,497	observations	`getDfSummary(dfBrookHeat)`	OBSERVED E-2B1FA53FA7
Filtered model population	population stage	67,495	observations	`dfmodel = buildFeatures(dfBrookHeat) dfmodel = dfmodel[['Incident Zip','Day of Week','Day of Month','Month','descriptorAPARTMENT ONLY', 'descriptorENTIRE BUILDING','ResolutionTime']] dfmodel.describe()`	OBSERVED E-2B1FA53FA7
Filtered model population	population stage	67,485	observations	`bins = [0,2,6,28] groupnames = [0,1,2] dfmodel['categories'] = pd.cut(dfmodel['ResolutionTime'], bins,includelowest=True,labels=groupnames) dfmodel.describe()`	OBSERVED E-2B1FA53FA7
Training population	training	47,239	observations	`70/30 train/test split`	DERIVED E-2B1FA53FA7
Held-out evaluation population	evaluation	20,246	observations	`70/30 train/test split`	DERIVED E-2B1FA53FA7

Complete model-comparison tables

Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data

Results recorded for the same target and evaluation setup.

METRIC	LOGISTIC REGRESSION	DECISION TREE	RANDOM FOREST
Accuracy calculated from yes/no predictions	0.5780894991603279	0.621456090092	0.6248641707003852

Random Forest records the strongest value across all 1 comparable metrics shown for Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data. The comparison is still row-specific: the values describe different evaluation measures and should not be compared vertically as though they shared one scale.

Evidence E-2B1FA53FA7

Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data

Results recorded for the same target and evaluation setup.

METRIC	LOGISTIC REGRESSION	DECISION TREE	RANDOM FOREST
Accuracy calculated from yes/no predictions	0.5740887088807666	0.611478810629	0.6185913266818137

Random Forest records the strongest value across all 1 comparable metrics shown for Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data. The comparison is still row-specific: the values describe different evaluation measures and should not be compared vertically as though they shared one scale.

Evidence E-2B1FA53FA7

Prevalence and baseline context

No supported entries were reconstructed for this table.

Additional evaluation diagnostics

TARGET	METHOD	DIAGNOSTIC	PHASE	PERSISTED VALUE	EVIDENCE
Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data	Random Forest	confusion matrix	test	`[[0.55115806,0.41639094,0.03245101],[0.22262401,0.73546592,0.04191006],[0.22333125,0.53462258,0.24204616]]`	E-2B1FA53FA7
Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data	Decision Tree	confusion matrix	test	`[[0.60244337,0.37223212,0.02532451],[0.2676866,0.69568846,0.03662494],[0.26699938,0.51777916,0.21522146]]`	E-2B1FA53FA7
Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data	Decision Tree	train validation	test	–	E-2B1FA53FA7
Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data	Decision Tree	train validation	test	–	E-2B1FA53FA7

Full feature-evidence sets

Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data

Logistic Regression · recorded inputs

Incident Zip	**Used**
Day of Week	**Used**
Day of Month	**Used**
Month	**Used**
Descriptor: Apartment Only	**Used**
Descriptor: Entire Building	**Used**

The recorded model input contains Incident Zip, Day of Week, Day of Month, and Month and 2 additional recorded inputs. The evidence establishes that these fields were used, but it does not provide a relative-importance ranking or establish causality.

The evidence establishes feature use, not relative importance or cause and effect.

Evidence E-2B1FA53FA7

Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data

Decision Tree · recorded inputs

Incident Zip	**Used**
Day of Week	**Used**
Day of Month	**Used**
Month	**Used**
Descriptor: Apartment Only	**Used**
Descriptor: Entire Building	**Used**

The recorded model input contains Incident Zip, Day of Week, Day of Month, and Month and 2 additional recorded inputs. The evidence establishes that these fields were used, but it does not provide a relative-importance ranking or establish causality.

The evidence establishes feature use, not relative importance or cause and effect.

Evidence E-2B1FA53FA7

Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data

Random Forest · recorded inputs

Incident Zip	**Used**
Day of Week	**Used**
Day of Month	**Used**
Month	**Used**
Descriptor: Apartment Only	**Used**
Descriptor: Entire Building	**Used**

The recorded model input contains Incident Zip, Day of Week, Day of Month, and Month and 2 additional recorded inputs. The evidence establishes that these fields were used, but it does not provide a relative-importance ranking or establish causality.

The evidence establishes feature use, not relative importance or cause and effect.

Evidence E-2B1FA53FA7

Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data

Logistic Regression · recorded inputs

Incident Zip	**Used**
Day of Week	**Used**
Day of Month	**Used**
Month	**Used**

The recorded model input contains Incident Zip, Day of Week, Day of Month, and Month. The evidence establishes that these fields were used, but it does not provide a relative-importance ranking or establish causality.

The evidence establishes feature use, not relative importance or cause and effect.

Evidence E-2B1FA53FA7

Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data

Decision Tree · recorded inputs

Incident Zip	**Used**
Day of Week	**Used**
Day of Month	**Used**
Month	**Used**

The recorded model input contains Incident Zip, Day of Week, Day of Month, and Month. The evidence establishes that these fields were used, but it does not provide a relative-importance ranking or establish causality.

The evidence establishes feature use, not relative importance or cause and effect.

Evidence E-2B1FA53FA7

Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data

Random Forest · recorded inputs

Incident Zip	**Used**
Day of Week	**Used**
Day of Month	**Used**
Month	**Used**

The recorded model input contains Incident Zip, Day of Week, Day of Month, and Month. The evidence establishes that these fields were used, but it does not provide a relative-importance ranking or establish causality.

The evidence establishes feature use, not relative importance or cause and effect.

Evidence E-2B1FA53FA7

Tuning and search stages

No supported entries were reconstructed for this table.

Model variants

TARGET	METHOD	ESTIMATOR	ROLES	PARAMETERS	SOURCE	EVIDENCE
Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data	Logistic Regression	LogisticRegression	untuned	`C=1e30`	NYC311.ipynb#cell-49	E-2B1FA53FA7
Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data	Decision Tree	DecisionTreeClassifier	untuned, held out untuned, export producing	`criterion='entropy'`	NYC311.ipynb#cell-52	E-2B1FA53FA7
Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data	Random Forest	RandomForestClassifier	untuned, held out untuned, export producing	`n_estimators=100`	NYC311.ipynb#cell-54	E-2B1FA53FA7
Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data	Logistic Regression	LogisticRegression	untuned	`C=1e30`	NYC311.ipynb#cell-69	E-2B1FA53FA7
Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data	Decision Tree	DecisionTreeClassifier	untuned	`criterion='entropy'`	NYC311.ipynb#cell-71	E-2B1FA53FA7
Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data	Random Forest	RandomForestClassifier	untuned, held out untuned, export producing	`n_estimators=100`	NYC311.ipynb#cell-73	E-2B1FA53FA7

Selection statements

STATEMENT	TARGET	METHOD	METRIC	VALUE	STATUS	EVIDENCE
best final on metric	Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data	Random Forest	accuracy	0.624864	established	E-2B1FA53FA7
best final on metric	Let's predict the resolution time for Heat issues in Brooklyn given the 2015 data	Random Forest	accuracy	0.618591	established	E-2B1FA53FA7
project selected model	Target not established	–	–	–	unresolved	

Output relationships

No supported entries were reconstructed for this table.

Scoped limitations

No supported entries were reconstructed for this table.

Checks performed

CHECK	QUESTION	RESULT	IMPLEMENTATION	SUMMARY	EVIDENCE
CHK-CURRENT-PURPOSE-TRACE	Current purpose source	Not detected	implemented	Current purpose source is resolved to the inspected evidence.	E-B335630551 · E-B9CC546808
CHK-PROJECT-CONTEXT-COVERAGE	Material project context	Not detected	implemented	Material project context is resolved to the inspected evidence.	E-B335630551 · E-B9CC546808 · E-2B1FA53FA7
CHK-SOURCE-INVENTORY-COVERAGE	Authorised source inventory	Not detected	implemented	Authorised source inventory is resolved to the inspected evidence.	E-B335630551 · E-B9CC546808
CHK-TYPED-FOCAL-RESULT-ORIGIN	Typed focal-result origin	Not detected	implemented graph native	The focal result resolves through compatible evidence-bound target, run, evaluation, metric, and population objects.	E-2B1FA53FA7
CHK-TARGET-CONSTRUCTION-ANCESTRY	Complete target construction ancestry	Not enough evidence	implemented graph native	The target construction ancestry contains an unresolved step, edge, leaf, or evidence binding.	E-2B1FA53FA7
CHK-POPULATION-LINEAGE-COMPATIBILITY	Population lineage compatibility	Not enough evidence	implemented graph native	The population lineage is missing an original, training or evaluation population, a connecting relationship, or its supporting evidence.	E-2B1FA53FA7
CHK-EVALUATION-INTENDED-POPULATION-ALIGNMENT	Evaluation versus intended prediction population	Not applicable	implemented graph native	No population-specific intended prediction or decision population is represented.	
CHK-EVALUATION-PREPROCESSOR-FIT-SCOPE	Evaluation preprocessing fit scope	Not applicable	implemented graph native	No fitted preprocessing transformation is represented in the available project record.	
CHK-METRIC-COMPUTATION-CLAIM-COMPATIBILITY	Metric computation and interpretation compatibility	Not applicable	implemented graph native	No material ranking or curve metric is reconstructed in the available project record.	
CHK-POSITIVE-CLASS-DETECTION-DEGENERACY	Positive-class detection behaviour	Not applicable	implemented graph native	No final binary class-decision diagnostic is represented.	
CHK-TERMINAL-SEARCH-FINAL-CORRESPONDENCE	Terminal search and final estimator correspondence	Not applicable	implemented graph native	No parameter-search stage requiring a search-to-final comparison is reconstructed in the available project record.	
CHK-OUTPUT-COLUMN-CONSISTENCY	Output column consistency	Not applicable	implemented graph native	No prediction output-production statement is represented.	
CHK-ANALYTICAL-ROLE-ALIGNMENT	Analytical model-role alignment	Not applicable	implemented graph native	Fewer than two material analytical roles are resolved for this target.	E-2B1FA53FA7
CHK-REPRODUCTION-INPUT-COVERAGE-V2	Result reproduction inputs	Not enough evidence	implemented graph native	One or more independent reproduction-input categories are unresolved in the available project record.	E-2B1FA53FA7
CHK-MATERIAL-ANALYTICAL-AREA-COVERAGE	Material analytical area coverage: Main result	Not detected	implemented analytical area native	Main result answers its defined supervisory question with connected evidence.	E-2B1FA53FA7 · E-B9CC546808

CHECK	QUESTION	RESULT	IMPLEMENTATION	SUMMARY	EVIDENCE
CHK-MATERIAL-ANALYTICAL-AREA-COVERAGE	Material analytical area coverage: Model comparison	Not detected	implemented analytical area native	Model comparison answers its defined supervisory question with connected evidence.	E-2B1FA53FA7 · E-B9CC546808
CHK-MATERIAL-ANALYTICAL-AREA-COVERAGE	Material analytical area coverage: Feature / driver evidence	Not enough evidence	implemented analytical area native	Bóveda found enough evidence to establish that feature / driver evidence is applicable, but not enough to answer the supervisory question that the section represents.	E-2B1FA53FA7 · E-B9CC546808
CHK-MATERIAL-ANALYTICAL-AREA-COVERAGE	Material analytical area coverage: Evaluation behaviour	Not detected	implemented analytical area native	Evaluation behaviour answers its defined supervisory question with connected evidence.	E-2B1FA53FA7 · E-B9CC546808
CHK-MATERIAL-ANALYTICAL-AREA-COVERAGE	Material analytical area coverage: Population and sample lineage	Not detected	implemented analytical area native	Population and sample lineage answers its defined supervisory question with connected evidence.	E-2B1FA53FA7 · E-B9CC546808
CHK-MATERIAL-ANALYTICAL-AREA-COVERAGE	Material analytical area coverage: Missing data before preparation	Not applicable	implemented analytical area native	Missing data before preparation is not demonstrably applicable to the reconstructed main-target story.	E-2B1FA53FA7 · E-B9CC546808
CHK-MATERIAL-EVIDENCE-ABSENCE	Material evidence absence	Condition found	implemented finding native	Material evidence absence prevents meaningful supervisory assessment of feature / driver evidence.	E-2B1FA53FA7 · E-B9CC546808

Complete History projection

DATE	TYPE	EVENT	DESCRIPTION	ROLE	EVIDENCE
2015	data period	Project data period reconstructed	The prepared service-request analysis and resolution model used data from 2015.	Material	E-2B1FA53FA7
2015	evaluation	Primary evaluation recorded	Persisted project evidence records Test accuracy 0.62486417070038525 for Three resolution-time classes for Brooklyn heat complaints using Random Forest Classifier with 100 trees.	Material	E-B9CC546808 · E-2B1FA53FA7
26 Aug 2026, 11:58 UTC	boveda audit	Bóveda audit completed	Bóveda completed audit AUD-8663C7D0D4 and stored the current canonical Project Record. 2 reconstruction attempts were recorded; 1 did not pass validation and 1 triggered a retry.	Material	

The PDF retains material History rows. 15 supporting project changes remain available in the complete HTML report.

Technical provenance

ANALYTICAL GRAPH SCHEMA	DERIVATION
boveda-analytical-object-graph-0.13.2	`"deterministic_persisted_evidence_v0.13.2"`

COMPLETENESS	PROVIDER CALLS USED BY THIS ANALYTICAL PROJECTION
`{"targets":"available","populations":"available","model_runs":"available","searches":"unavailable","metrics":"available","diagnostics":"available","prevalence":"unavailable","outputs":"unavailable","evidence_termination":"complete"}`	0

APPENDIX B

Evidence appendix

A stable index linking material report claims to source evidence retained in the Project Record.

Evidence Index

EVIDENCE TOKEN	SOURCE	SOURCE LOCATOR	EPISTEMIC STATUS	SHORT DESCRIPTION
E-2B1FA53FA7	NYC311.ipynb	NYC311.ipynb	OBSERVED	[CODE FOR OUTPUT] df_perfect = prepareData(df) df_perfect.shape [PERSISTED OUTPUT] (2099575, 57) [CODE FOR OUTPUT] df_Brooklyn.shape [PERSISTED OUTPUT] (654694, 57) [CODE FOR OUTPUT] from...
E-B335630551	README.md	README.md	OBSERVED	# Analysis of NYC-311 Service Request Data Background: 311 Service Requests encompass all non-emergency requests from the city, including but not limited to noise complaints, air quality...
E-B9CC546808	Project_Report_GIT.pdf	Project_Report_GIT.pdf	OBSERVED	Exploring 311 Service Request Data Problem Statement and Background: Background: 311 Service Requests encompass all non-emergency requests from the city, including but not limited to nois...
E-SYSTEM-PATH	.	.	OBSERVED	Imported directory name: R3_public-services_nyc-311-resolution

The PDF index includes evidence linked to material report content. 16 additional supporting History entries remain indexed and expandable in the complete HTML report.

Raw evidence excerpts

Raw excerpts are separated from the client narrative. Browser readers may expand every retained excerpt, and longer material is clearly marked as partial in print. The PDF includes bounded excerpts for evidence linked to material report content; supporting source-control History remains complete in the self-contained HTML report.

E-2B1FA53FA7 · NYC311.ipynb

Epistemic status: OBSERVED · **Kind:** notebook

This is the complete excerpt retained in the Project Record. Source collection bounded this excerpt; the source may contain additional content. The HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

```
[CODE FOR OUTPUT]
df_perfect = prepareData(df)
df_perfect.shape

[PERSISTED OUTPUT]
(2099575, 57)

[CODE FOR OUTPUT]
df_Brooklyn.shape

[PERSISTED OUTPUT]
(654694, 57)

[CODE FOR OUTPUT]
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

X_train, X_test, y_train, y_test = splitter(df_model,df_model['categories'])
X1 = X_train.drop(['Resolution_Time','categories'],1)
Y1 = y_train
X2 = X_test.drop(['Resolution_Time','categories'],1)
Y2 = y_test
log=LogisticRegression(C=1e3

[Middle of retained excerpt omitted from the print edition.]

.) (ZONE)                2010-02-09       2100
217496         KINGS (BROOKLYN) (ZONE)               2010-02-09       2200

           EVENT_TYPE MAGNITUDE TOR_F_SCALE  DEATHS_DI
```

E-B335630551 · README.md

Epistemic status: OBSERVED · **Kind:** documentation

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

```
# Analysis of NYC-311 Service Request Data
Background:

311 Service Requests encompass all non-emergency requests from the city, including but not limited to noise
complaints, air quality issues and reports of unsanitary conditions etc. The 311 calls in New York City (NYC)
are publicly available.

311 service request dataset is available at https://data.cityofnewyork.us/Social-Services/311- Service-Requests-
from-2010-to-Present/erm2-nwe9

Problem Statements and Motivation:

Analysis of 311 calls can be of great us

[Middle of retained excerpt omitted from the print edition.]

oss NYC](https://github.com/ayush159/NYC-311/blob/master/graphs/air_quality.png)

![Response Time Comparison](https://github.com/ayush159/NYC-311/blob/master/graphs/complaints.png)
```

E-B9CC546808 · Project_Report_GIT.pdf

Epistemic status: OBSERVED · **Kind:** report

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

Print boundary: this edition includes the beginning and end of the retained excerpt and clearly marks the omitted middle. The self-contained HTML report retains the complete Project Record excerpt.

```
Exploring 311 Service Request Data

Problem Statement and Background:

Background:

311 Service Requests encompass all non-emergency requests from the city, including but not
limited to noise complaints, air quality issues and reports of unsanitary conditions etc. The
311 calls in New York City (NYC) are publicly available.

311 service request dataset is available  at https://data.cityofnewyork.us/Social-Services/311-
Service-Requests-from-2010-to-Present/erm2-nwe9
This dataset comprises of all calls made to 311 f

[Middle of retained excerpt omitted from the print edition.]

so filtered on many levels to
reduce the data size.
2. Assumed that all the complaints were in closed state.
3. Assumed that impact of storm lasted only till the duration of storm.
```

E-SYSTEM-PATH · .

Epistemic status: OBSERVED · **Kind:** project identity

This is the complete excerpt retained in the Project Record; the HTML report applies no further truncation.

```
Imported directory name: R3_public-services_nyc-311-resolution
```

APPENDIX C

Bóveda processing metadata

Technical audit diagnostics are retained as a subordinate appendix and do not compete with the project narrative.

TELEMETRY AVAILABILITY complete	AUDIT DURATION 68.7 s
RECONSTRUCTION RESULT accepted	ATTEMPTS / PROVIDER CALLS 2 / 2
TOTAL RECORDED TOKENS 34,189	PAID MODEL ACTIVITY DURING STORED AUDIT Yes
LOCALISED REPAIR accepted · record.samples.source_data, record.results_evaluation.establishes, record.data.period	

Attempt 1 · full reconstruction

rejected

Duration	56.4 s
Recorded tokens	20,517
Validation findings	• record.samples.source_data: numeric value 16 is not grounded in cited evidence • record.results_evaluation.establishes: numeric value 62.486% is not grounded in cited evidence • record.data.period: numeric value 311 is not grounded in cited evidence
Failure stage	grounding · record.samples.source_data: numeric value 16 is not grounded in cited evidence · record.results_evaluation.establishes: numeric value 62.486% is not grounded in cited evidence · record.data.period: numeric value 311 is not grounded in cited evidence
Retry	The rejected fields were supplied to the next attempt as corrections for a localised repair.

Attempt 2 · localized repair

accepted

Duration	12.2 s
Recorded tokens	13,672
Validation findings	None recorded